

Ô (O-Hat) Cross-Domain Classification Spectrum v2.0

A substrate-independent structural measurement framework validated across 12+ domains.

Abstract

Ô is a three-number operator (curl, helicity, balance) that measures structural properties of any time series — regardless of what the numbers represent. It classifies dynamical systems into a spectrum determined by baseline background activity density, not outbreak intensity. After cross-domain validation across climate, epidemiology, seismology, space weather, meteorology, AI safety, and finance, we present the v2.0 classification spectrum with seven system types, the balance dual-role hypothesis, and a 3-6-9 catastrophe-theoretic foundation.

1. The Three Metrics

Metric	Formula	What It Captures	Physical Meaning
curl	$\Sigma(\Delta x)$	Signed gradient sum	Directional acceleration — is the system moving toward or away from equilibrium?
helicity	$\Sigma($	Δx	$)$
balance		skew_outbreak	/

Key insight: All three are substrate-independent. They work because skewness and gradients exist in any ordered sequence, regardless of domain semantics.

2. The Seven System Types (v2.0)

2.1 Primary Spectrum (Balance-Ratio Based)

Balance Ratio	System Type	Characteristics	Verified Examples
> 20×	SUPER-PULSE		

Balance Ratio	System Type	Characteristics	Verified Examples
		Near-perfect Gaussian baseline + moderate outbreak. Requires extreme baseline purity.	1989 Quebec geomagnetic storm (40.3×, baseline)
5-15×	PULSE	Sparse baseline (mostly zero/minimal events) → large contrast with burst	Earthquakes M5+ (9.2×)
1.5-4×	CONTINUOUS FLOW	Baseline has regular activity → moderate but significant contrast with outbreak	ENSO (2.2×), COVID-19 (2.5×), Typhoon storm intensity (2.6×)
< 1.5×	REGIME SATURATION	System too strongly forced by external periodic driver → outbreak is more regular than baseline	Typhoon seasons (0.4×), some river flood regimes

2.2 Secondary Types (Structure-Based, H_v3 Sensitive)

These types require the expanded H_v3 metric (helicity of the outbreak window) in addition to balance ratio.

Type	Signature	Mechanism	Verified Examples
VOLATILITY SHIFT (R_v)	Balance < 1, helicity ↑	Variance changes without mean shift	S&P 500 crash regimes
COLLAPSE (Type X)	H_v3 ↓ (structure destruction)	System loses internal organization during event	Geomagnetic storms (2003 Halloween Storm: H_v3=0.42× of baseline), microservice cascading failures
PERTURBATION	H_v3 ↑ (structure emergence)	Event introduces NEW structure not present in baseline	ICS cyberattacks (H_v3 increases), LLM jailbreak prompts, adversarial attacks

2.3 Type X vs Perturbation — The Critical Distinction

The same balance ratio can mask two opposite dynamics:

Type X:	baseline structured → event chaotic	(H_v3 ↓)
Perturbation:	baseline simple → event complex	(H_v3 ↑)

This distinction emerges naturally from the 3-6-9 cusp catastrophe framework (see §4).

3. Scale-Dependent Transitions

Some systems exhibit **different types at different scales** — $\hat{O}$ detects this as a feature, not a bug.

System	Micro Scale	Macro Scale	Mechanism
Typhoons	CONTINUOUS FLOW (2.6×, individual storm intensity)	REGIME SATURATION (0.4×, seasonal cycle)	External seasonal forcing dominates macro scale
LLM Attention	PERTURBATION (jailbreak prompts)	CONTINUOUS FLOW (normal use)	Attention pattern shift is local, not global

Implication: $\hat{O}$ is a meta-metric that reveals scale-dependent structure. The "correct" classification depends on the scale you care about.

4. Theoretical Foundation: 3-6-9 Cusp Catastrophe

4.1 $\Sigma(t)$ Formulation

The framework maps to a three-variable state vector:

$$\Sigma(t) = (x, \dot{x}, V)$$

Where: - $\mathbf{x}$ = system state (curl maps to this) - $\dot{\mathbf{x}}$ = rate of change (helicity maps to this)
- $\mathbf{V}$ = potential energy / stress (balance maps to this)

4.2 Cusp Catastrophe Geometry

The cusp catastrophe surface has three behavioral modes: - **9** (stable equilibrium) — system on lower sheet - **6** (bifurcation zone) — system approaching fold curve - **3** (catastrophe jump) — system crosses fold, jumps to upper sheet

4.3 Directional Reversal (9 → 3 → 6)

A key prediction: after a catastrophe jump (State 3), the system does NOT return along the same path. It must traverse back (State 6) before restabilizing (State 9). This directionality is visible in the curl sign reversal between Type X and Perturbation events.

5. The Balance Dual-Role Hypothesis

Balance ratio serves two diagnostic functions simultaneously:

Role	What It Measures	Example
Outbreak Detector	How different is the outbreak from baseline?	Brazil COVID: high outbreak skew → high balance
Containment Dynamics	Does the system return to equilibrium between events?	UK COVID: poor containment → elevated baseline skew → LOW balance despite severe outbreak

This was discovered during Brazil vs UK COVID-19 comparison — two countries with similarly severe outbreaks showed different balance ratios because UK's poor containment elevated the baseline, making the outbreak appear "less unusual" structurally.

6. The Curl × Helicity 2×2 Matrix

Crossing curl (directional) with helicity (magnitude) yields four behavioral quadrants:

	Low Helicity	High Helicity
Positive Curl	Gentle drift away from equilibrium	Accelerating structural divergence
Negative Curl	Gentle return toward equilibrium	Accelerating structural collapse

This matrix was independently identified by DeepSeek's analysis of the $\hat{O}$ framework and confirmed by reanalysis of LLM attention-layer data.

7. Complete Domain Coverage

#	Domain	System Type	Balance Ratio	Key Finding	Confidence
1	ENSO (climate)	CONTINUOUS FLOW	~2.2×	92% of El Niño events detected; baseline activity from regular ENSO cycle	☐ Verified
2	COVID-19 (epidemiology)	CONTINUOUS FLOW	~2.5× (Brazil), varies by country	Balance reveals containment quality, not just outbreak severity	☐ Verified
3	Earthquakes M5+ (seismology)	PULSE	9.2×	Sparse inter-event baseline creates high contrast;	☐ Verified

#	Domain	System Type	Balance Ratio	Key Finding	Confidence
				window-size sensitive	
4	1989 Quebec Storm (space weather)	SUPER-PULSE	40.3×	Baseline	skew
5	2003 Halloween Storm (space weather)	COLLAPSE (Type X)	N/A	H_v3=0.42× — structure destruction during event	☐ Verified
6	Typhoons — storm scale (meteorology)	CONTINUOUS FLOW	2.6×	Individual storm intensity follows same pattern as ENSO/COVID	☐ Verified
7	Typhoons — seasonal scale (meteorology)	REGIME SATURATION	0.4×	Seasonal forcing overwhelms anomaly signal at macro scale	☐ Verified
8	LLM Hidden States (AI safety)	Mixed	Varies	Type X for geomagnetic-storm analogy; Perturbation for jailbreaks	☐ Verified
9	LLM Attention Layers (AI safety)	Mixed	Varies	Curl best discriminator for JB vs LM; seq length is dominant confound	☐ Verified
10	ICS Cyberattacks (security)	PERTURBATION	Varies	H_v3 ↑ — attack introduces structure not present in baseline	☐ Verified
11	S&P 500 (finance)	VOLATILITY SHIFT	< 1	Variance change without mean shift; balance < 1, helicity ↑	△ Preliminary
12	Microservice Failures (systems)	COLLAPSE (Type X)	Varies	Cascading failure destroys internal structure	△ Preliminary

Untested Domains (Priority Queue)

Domain	Expected Type	Rationale
Volcanic eruptions	PULSE	Sparse events, similar to earthquakes
River floods	Scale-dependent	Like typhoons — seasonal forcing at macro, pulse at event scale
Stellar light curves	Unknown	Potentially novel transient structure
Social media virality	PULSE or SUPER-PULSE	Sparse events with massive contrast
LLM perplexity under attack	PERTURBATION	Attention-layer findings suggest $H_{v3} \uparrow$

8. Known Limitations

See `docs/known-limitations.md` for detailed analysis. Key failure modes:

1. **Crime rate data** — long-term secular trends collapse baseline/outbreak distinction
2. **Random walks** — no inherent structure to measure
3. **Multi-modal distributions** — skewness assumption breaks
4. **Very short series (< 50 points)** — auto-parameterization fails
5. **Heavily smoothed data** — gradient metrics suppressed
6. **Strong external periodic forcing** — $\hat{O}$ detects the clock, not the anomaly

9. The Cross-Domain $\sim 2.2\times$ Anomaly

Observation: Three domains converge on balance ratios near $2.2\times$: - ENSO: $2.2\times$ - COVID-19 (Brazil): $2.5\times$ - Typhoon storm intensity: $2.6\times$

Hypothesis: $\sim 2.2\times$ represents a universal "baseline-active outbreak" ratio — systems where the background never goes to zero but outbreaks still produce detectable structural change.

Falsification strategy: Find a domain with continuous baseline activity AND a balance ratio outside the $1.5\text{--}4\times$ range.

10. References

Primary (Zenodo, peer-accessible)

#	Title	DOI	Date
1	A Simple Way to Measure El Niño	10.5281/zenodo.21415286	2026-07-17
2	Ô Cross-Domain Synthesis Note	10.5281/zenodo.21416188	2026-07-17
3	COVID-19 Cross-Domain Validation	10.5281/zenodo.21416411	2026-07-17
4	H-Detector v0.4 — Cross-Domain Structural Detection	10.5281/zenodo.21383080	2026-07-15

Software

Resource	URL
Ô (O-Hat) GitHub Repository	https://github.com/MMDR10/o-hat
Colab Demo Notebook	Included in <code>examples/o_hat_demo.ipynb</code>

Theoretical Foundation (External)

- Zeeman, E.C. (1976). "Catastrophe Theory." Scientific American, 234(4), 65-83.
- Gilmore, R. (1981). Catastrophe Theory for Scientists and Engineers. Wiley.

Appendix A: Evidence-Level Notation

Label	Meaning	Criterion
☐ Verified	High confidence	≥ 2 independent sources, consistent conclusions
⚠ Preliminary	Medium confidence	Single-source or limited verification
☐ Untested	No confidence	No data yet

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